Limiting cases for Finte term $(\Lambda \to 0,\ \vee \to \infty)$

1 Group I (with c.c.)

Reality. Each term here is $T = \int e^{-ik\cdot\Delta}\, c\,\mathcal{K}\,\mathcal{O}$ with a real kernel $\mathcal{K}$, and the c.c. is added. For classical ρ: if c is real, $T + T^\dagger = 2c \int \cos(k\cdot\Delta)\,\mathcal{K}\,\mathcal{O}$; if $c = i\alpha$ with α real, $T + T^\dagger = 2\alpha \int \sin(k\cdot\Delta)\,\mathcal{K}\,\mathcal{O}$, since $i\,e^{-ik\cdot\Delta} + \text{c.c.} = 2\sin(k\cdot\Delta)$. So every explicit i disappears. The only logarithms with possibly negative arguments are in Row III; they are treated there.

1.1 Row I

NA

1.2 Row II

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)} \int d^2p\, d^2q \int_{x,y} e^{iq\cdot(w-y)}\, e^{ip\cdot(y-x)}\, \frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{q}^2}\, R^i(\mathbf{q},\mathbf{p})$$
$$\times \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big] \tag{1.1}$$

where

$$R^i = -\,2q^i\log\left(\frac{p^2}{q^2}\right) + \frac{(2p^2 - 2\,q\cdot p + q^2)\,q^i + q^2p^i}{(q-p)^2}\,\log\left[\frac{q^2}{(q-p)^2}\right]. \tag{1.2}$$

The integration momentum is renamed $k \to q$, because it clashed with the external k. Every term of R^i is odd under $(q,p) \to (-q,-p)$, so the double Fourier integral is $i\times$(real). With the c.c. this row is therefore real (sin).

1.3 Row III

We write it in three parts:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left[\frac{(x-z)^2}{(x-z)^2-(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{-(x-z)^2}{(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{1.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$

$$\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\,\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{1.4}$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right]\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(w)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.6}
$$

where:

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan\left[\frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] + \arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] \right\} = \frac{\theta_w}{\left|(x-w)\times(z-w)\right|}, \qquad \theta_w = \arccos\frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|} \in [0,\pi] \tag{1.7}$$

Here θ_w is the angle of the triangle (x,z,w) at w. This uses $(x-z)^2-(x-w)\cdot(x-z) = -(x-z)\cdot(z-w)$ and $|(x-w)\times(x-z)| = |(z-w)\times(x-z)|$.

Correction to the last term of N_{3b}. At finite Λ this term carries $\log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2}$, which tends to $2\log\frac{k^+-\Lambda}{\Lambda}$ as $z\to w$. With the kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ it is therefore *non-integrable* at $z\to w$, even at finite Λ. Its $1/(z-w)^2$ singularity was cancelled *exactly*, for every Λ, by the original $-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\frac{k^+-\Lambda}{\Lambda}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ term of N_{3a}. That term was replaced, by the JIMWLK-consistency requirement verified at operator level, with the kernel $\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2}$ and $U^{bc}(z)\to U^{bc}(w)$. Both terms come from the same δ-piece of the p^+ integrand, so the same replacement is applied here (in red above). The corrected term is integrable at $z\to w$: it behaves as $\log(1/|z-w|)/|z-w|$, and its angle average vanishes. This was checked numerically. Without the correction, $r^2\times$(angle average) tends to a nonzero constant, $\propto\log(k^+/\Lambda)$, as $r=|z-w|\to 0$.

Row III simplified: the negative-argument logarithms

Let $u=\frac{(x-z)^2}{(x-w)^2}$. The logarithms come from the energy denominator of $\bar{\mathbb{B}}_3^\dagger$, $p^+(x-z)^2-k^+(x-w)^2-i0$, i.e. they are on the *principal branch*, $\log(-A)=\log A+i\pi$ for $A>0$. No absolute values are needed. With θ the step function:

$$\log\frac{(x-z)^2}{(x-z)^2-(x-w)^2} = \log\frac{u}{|1-u|}+i\pi\,\theta(1-u), \qquad \log\frac{(x-w)^2-(x-z)^2}{(x-w)^2} = \log|1-u|+i\pi\,\theta(u-1),$$
$$\log\frac{-(x-z)^2}{(x-w)^2-(x-z)^2} = \log\frac{u}{|1-u|}+i\pi\,\theta(1-u), \qquad \log\frac{-(x-z)^2}{(x-w)^2} = \log u+i\pi. \tag{1.8}$$

The pairs that appear together add up to the same function for all $u>0$:

$$\log\frac{(x-z)^2}{(x-z)^2-(x-w)^2}+\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2} = \log\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}+\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2}$$
$$= \log\frac{-(x-z)^2}{(x-w)^2} = \log u+i\pi. \tag{1.9}$$

- In N_1, the first two terms have the same kernel and give the first pair.
- The six terms of N_2 ($\log\log\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}$) and the first six terms of N_{3a} ($\log\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2}$) contain the same six tensors with the same coefficients, so they give the second pair.

All logarithmic terms of $N_1+N_2+N_{3a}$ therefore multiply the single factor $\log u+i\pi$. With $X=x-w$, $Z=x-z$, $W=z-w=X-Z$, their tensor coefficient collapses to

$$\frac{X^2+Z^2}{(X^2-Z^2)^2}\delta^{ij}-\frac{2X^iZ^j}{X^2Z^2}+\frac{2}{W^2(X^2-Z^2)}\left[\frac{(W\cdot Z)X^2-(W\cdot X)Z^2}{X^2-Z^2}\delta^{ij}-W^iX^j+\frac{Z^2}{X^2}X^iW^j+\frac{X^2}{Z^2}W^iZ^j-Z^iW^j\right] = 2\,\mathcal{K}^{ij}_{\rm IV}. \tag{1.10}$$

For the δ^{ij} part, $\frac{(X^2+Z^2)W^2+2(X\cdot Z)(X^2+Z^2)-4X^2Z^2}{W^2(X^2-Z^2)^2}=\frac{1}{W^2}$. For the rest, the difference of the two sides is $-W^i(X-Z)^j+(X-Z)^iW^j=0$. Here

$$\mathcal{K}^{ij}_{\rm IV} = \frac{\delta^{ij}}{2(z-w)^2}+\frac{(z-w)^i(x-z)^j}{(z-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2} \tag{1.11}$$

is the Row IV kernel. All the $\frac{1}{(x-z)^2-(x-w)^2}$ poles cancel. This was checked numerically from the terms above, for $u<1$ and $u>1$, to 10^{-12}. Hence, with $X_\perp = X - \frac{X\cdot Z}{Z^2}Z$ and $Z_\perp = Z - \frac{X\cdot Z}{X^2}X$, Row III $= N_1 + N_2 + N_{3a} + N_{3b}$ becomes

$$\begin{aligned}
\left.\frac{d^3N_{\rm NLO}}{d^3k}\right|_{\rm III} &= \frac{1}{(2\pi)^3}\,\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\Bigg\{\frac{1}{(x'-w')^2(y-z)^2}\Bigg[2\log\frac{(x-z)^2}{(x-w)^2}\,(x'-w')^i(y-z)^j\,\mathcal{K}^{ij}_{\rm IV}\\
&+\frac{2\,\theta_w}{|(x-w)\times(z-w)|\,(z-w)^2}\Big((x'-w')\cdot(z-w)\;(y-z)\cdot X_\perp+(x'-w')\cdot Z_\perp\;(y-z)\cdot(z-w)\Big)\\
&+\frac{1}{(z-w)^2}\log\frac{(z-w)^2}{(x-w)^2}\Big(\frac{(x'-w')\cdot(x-w)\;(y-z)\cdot(z-w)}{(x-w)^2}-\frac{(x'-w')\cdot(z-w)\;(y-z)\cdot(x-z)}{(x-z)^2}\Big)\Bigg]\\
&\times\Big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y)-U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big]\\
&-\log\frac{(z-w)^2}{(x-w)^2}\,\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2(y-w)^2(x-z)^2(w-z)^2}\\
&\times\Big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(w)\rho^e(x)\rho^c(y)-U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big]\Bigg\}\\
&+\frac{d^3N_{i\pi}}{d^3k},
\end{aligned}\tag{1.12}$$

where the $i\pi$ of $\log u + i\pi$, multiplied by the overall i, gives a *real* term:

$$\begin{aligned}
\frac{d^3N_{i\pi}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4 f^{abd}}{8\pi^4}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\,\mathcal{K}^{ij}_{\rm IV}\\
&\times\Big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y)-U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\Big]\,.
\end{aligned}\tag{1.13}$$

Reality and finiteness.

- The first expression has coefficient $i\times$real; with its c.c. it becomes $\frac{g^4 f^{abd}}{8\pi^5 k^+}\int\sin(k\cdot(w'-w))\{\cdots\}$.
- $N_{i\pi}$ has a real coefficient; with its c.c. it becomes $-\frac{g^4 f^{abd}}{4\pi^4 k^+}\int\cos(k\cdot(w'-w))\{\cdots\}$.
- No logarithm of a negative number is left. The real parts are integrable everywhere.
- The only singular piece is the $\frac{\delta^{ij}}{2(z-w)^2}$ part of $\mathcal{K}_{\rm IV}$ in $N_{i\pi}$ at $z\to w$. There the colour factor becomes $f^{abd}X^a(w')\cdot Y^b(w)D^d$, which is antisymmetric under $w\leftrightarrow w'$. It multiplies the symmetric $\cos(k\cdot(w'-w))$, so it cancels exactly against the c.c.: it is a Coulomb phase.

1.4 Row IV

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times\\
&\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}\tag{1.14}$$

Positive logarithm; with the c.c. this row is $\frac{1}{(2\pi)^3}\frac{g^4}{4\pi^5}\frac{1}{k^+}\int\sin(k\cdot(w'-w))\{\cdots\}$. Its kernel $\mathcal{K}_{\rm IV}$ is the same one that appears in Row III.

1.5 Row V

First part

$$\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\\
&\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]
\end{aligned}\tag{1.15}$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.16}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{rest}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}\left\{e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\right.\\
&\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}-\left[\frac{k^+}{k^+-p^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.\\
&+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2\,(x-z)^2}\right]\\
&\left.\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\right.\\
&\left.-e^{-ik\cdot(w'-x)}\,\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}-2\right]\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\,\frac{(y-z)\cdot(y-x)}{(y-z)^2\,(x-z)^2}\right\}\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.17}
$$

2 Group II

Reality. There is no c.c.; each row is hermitian by itself. Row I, first part: $N_{1b}=N_{1a}^\dagger$, so $N_{1a}+N_{1b}=-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int\sin(k\cdot(w'-w))\{\cdots\}$. Row I, second part ($N_{2a}$–$N_{2f}$): real coefficients and positive logarithms. Each kernel is even under $(x,w)\leftrightarrow(x',w')$, which gives $\cos(k\cdot(w'-w))$. The poles $\frac{1}{(x-z)^2-(x-w)^2}$ are removable, because the attached log $\frac{(x-z)^2}{(x-w)^2}$ vanishes there. Row IV is symmetric under $(x,y)\leftrightarrow(x',y')$, which gives $\cos(k\cdot(y'-y)/\xi)$.

2.1 Row I

2.1.1 First part

$$\frac{d^3N_{1a\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\Bigg\}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{2.1}$$

$$\frac{d^3N_{1b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{(x-z)^2}{(x-w')^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{2.2}$$

2.1.2 Second part

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\left\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\right.$$
$$\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times$$
$$\left.\left[\log\left[\frac{(x-z)^2}{(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\right\}\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{2.3}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x-z)^2}{[(x'-w')^2(x-z)^2(x-w)^2-(x-w)^4(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2}+\frac{(x'-w')\cdot(x-w)(x'-z)\cdot(z-w)}{2(z-w)^2(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')(x-z)\cdot(z-w')}{2(x-z)^2(z-w')^2}+\frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-z)^2}\right]\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x)+N_cU^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-z)^2}{[(x'-w')^4(x-z)^2-(x-w)^2(x'-z)^2(x'-w')^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2}+\frac{(x'-w')\cdot(x-w)(x'-z)\cdot(z-w)}{2(z-w)^2(x'-z)^2}\right.$$
$$\left.+\frac{(x-w)\cdot(x'-w')(x-z)\cdot(z-w')}{2(x-z)^2(z-w')^2}+\frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-z)^2(x'-z)^2}\right]\times\log\left[\frac{(x'-z)^2}{(x'-w')^2}\right]$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x)+N_cU^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right] \tag{2.4}$$

$$\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')^2}{(x'-z)^2}\frac{1}{[(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2]}$$
$$\left[\frac{(x'-z)\cdot(x-z)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2}-\frac{(x'-z)\cdot(x-z)(x'-w')\cdot(z-w)}{2(z-w)^2(x'-w')^2}-\frac{(x-w)\cdot(z-w')(x-z)\cdot(x'-z)}{2(x-w)^2(z-w')^2}\right.$$
$$\left.+\frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-w)^2(x'-w')^2}\right]\times\log\left[\frac{(x'-z)^2}{(x'-w')^2}\right]$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x)+N_cU^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\frac{(x-w)^2}{(x-z)^2}\frac{1}{[(x'-w')^2(x-z)^2-(x-w)^2(x'-z)^2]}$$
$$\left[\frac{(x'-z)\cdot(x-z)(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2}-\frac{(x'-z)\cdot(x-z)(x'-w')\cdot(z-w)}{2(z-w)^2(x'-w')^2}-\frac{(x-w)\cdot(z-w')(x-z)\cdot(x'-z)}{2(x-w)^2(z-w')^2}\right.$$
$$\left.+\frac{(x-z)\cdot(x'-z)(x'-w')\cdot(x-w)}{4(x-w)^2(x'-w')^2}\right]\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]$$
$$\left[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x)+N_cU^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x)\right] \tag{2.5}$$

$$\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = +\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\log\left[\frac{(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.6}$$

$$\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$
$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$
$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.7}$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,[(x-z)^2-(x-w)^2]}\\
&\quad\frac{\big[(x'-z)^2-(x'-w')^2\big]}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&\quad -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&\quad +\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\quad\frac{\big[(x-z)^2-(x-w)^2\big]}{[(x'-z)^2-(x'-w')^2]\,[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\log\left[\frac{(x'-z)^2}{(x'-w')^2}\right]\times\\
&\quad\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&\quad - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{2.8}
$$

Second part simplified: $N_{2a}+\cdots+N_{2f}$ in one kernel

This is the $\vee\to\infty$ limit of the result in Other_terms (finite $\vee$). All pieces carry the same colour factor $\mathcal{C}_2$, the bracket of N_{2a} (it is common to all of them). $\mathcal{C}_2$ does not vanish at $z=w$ or $z=w'$, so the middle term of N_{2f}, $-\frac{d_\perp}{4}\frac{1}{(z-w)^2(z-w')^2}$, which is non-integrable at both points, has to cancel against the rest. Notation:

$$
\begin{aligned}
&X=x-w,\ Z=x-z,\ W=z-w,\ X'=x'-w',\ Z'=x'-z,\ W'=z-w', \qquad u=\frac{Z^2}{X^2},\ u'=\frac{Z'^2}{X'^2},\\
&\Delta\big[F(s)\big]\equiv\frac{F(u)-F(u')}{u-u'},\\
&V=\frac{W}{W^2}+\frac{Z}{2Z^2},\ P=\frac{W}{W^2}-\frac{X}{2X^2}\ \text{(and primed)},\\
&\mathcal{A}=(X\cdot P')(Z'\cdot V)+(X'\cdot P)(Z\cdot V'),\ \mathcal{B}=(X\cdot X')(V\cdot V'),\ \mathcal{C}=(Z\cdot Z')(P\cdot P'),\\
&\mathcal{D}=\frac{(Z'^2-X'^2)\,X\cdot V}{2W'^2}+\frac{(Z^2-X^2)\,X'\cdot V'}{2W^2}, \qquad \mathcal{E}=\frac{(Z'^2-X'^2)\,Z\cdot P}{2W'^2}+\frac{(Z^2-X^2)\,Z'\cdot P'}{2W^2}.
\end{aligned}
\tag{2.9}
$$

$\mathcal{A},\mathcal{B},\mathcal{C},\mathcal{D},\mathcal{E}$ are the brackets of $N_{2a},N_{2b},N_{2c},N_{2d},N_{2e}$. Then

$$
\begin{aligned}
\sum_{n=a}^{f}\frac{d^3N_{2n\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\left\{\frac{1}{(x-w)^2(x'-w')^2}\left[\mathcal{A}\,\Delta[\log s]-\mathcal{B}\,\Delta[s\log s]-\mathcal{C}\,\Delta\Big[\frac{\log s}{s}\Big]\right.\right.\\
&\quad\left.\left.+\mathcal{D}\,\Delta\Big[\frac{s\log s}{s-1}\Big]-\mathcal{E}\,\Delta\Big[\frac{\log s}{s-1}\Big]\right]-\frac{d_\perp}{4}\,\frac{\big[(x-z)^2-(x-w)^2\big]\big[(x'-z)^2-(x'-w')^2\big]}{(x-w)^2(x'-w')^2(z-w)^2(z-w')^2}\,\Delta\Big[\frac{s\log s-s+1}{(s-1)^2}\Big]\right\}\mathcal{C}_2.
\end{aligned}
\tag{2.10}
$$

The last term is all of N_{2f}, including its middle term, because $(u-1)(u'-1)\Delta\big[\frac{s\log s}{(s-1)^2}\big]+1=(u-1)(u'-1)\Delta\big[\frac{s\log s-s+1}{(s-1)^2}\big]$. All functions inside Δ are regular at $s=1$, so the poles at $u=u'$ and $u=1$, $u'=1$ are removable. Every term is at most $O(1/|z-w|)$ at $z\to w$ (and likewise at $z\to w'$), so the kernel is integrable. At $x\to z$ it is $O(\log|x-z|/|x-z|)$, also integrable. The kernel is symmetric under $(x,w)\leftrightarrow(x',w')$, so the result is real (cos). This was checked numerically against the twelve original pieces.

2.2 Row II

NA

2.3 Row III

NA

2.4 Row IV

$$\frac{d^3N_{\rm Finite}}{d^3k} = + \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]$$
$$\times\left[U^{ce'}(y') - U^{ce'}(x')\right]\left[U^{ce}(y) - U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x) \tag{2.11}$$

3 Group III (with c.c.)

Reality. All logarithms have positive arguments, and the denominators $p^+(x-z)^2 + k^+(x-w)^2$ are positive. Rows I, II, III (first part), V and VI have coefficient $i\times$real and give sin with the c.c.; Row III (second part) is real and gives cos.

3.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{d'b}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.1}$$

3.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(x-z)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]\times\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{c'b}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.2}$$

From here

3.3 Row III

3.3.1 First part

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee -k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee -k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left. - \frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{3.3}$$

3.3.2 Second part

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}$$
$$\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.$$
$$\left.\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\right.$$
$$-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]$$
$$\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times$$
$$\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}$$
$$\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times$$
$$\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^{e}(x)\right]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}$$
$$\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times$$
$$\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\quad(3.4)$$

Simplification of the third term of the second part. It has no p^+-dependent phase, so with $\vee\to\infty$

$$\int_0^\infty\frac{dp^+}{2\pi}\,\frac{(x-w)^2}{(x-z)^2(p^++k^+)\left[p^+(x-z)^2+k^+(x-w)^2\right]}=\frac{1}{2\pi k^+}\,\frac{(x-w)^2}{(x-z)^2\left[(x-z)^2-(x-w)^2\right]}\,\log\frac{(x-z)^2}{(x-w)^2}.\qquad(3.5)$$

So that term is $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\frac{(w'-z)^m(x'-z)^{k'}}{(w'-z)^2(x'-z)^2}\frac{(x-w)^2(x-z)^{k'}}{(x-z)^2[(x-z)^2-(x-w)^2]}\log\frac{(x-z)^2}{(x-w)^2}\left[\frac{(z-w)^m}{(z-w)^2}-\frac{(x-w)^m}{2(x-w)^2}\right]\times$(same colour). The pole at $(x-z)^2=(x-w)^2$ is removable. The kernel is integrable at $x\to z$, where it is $O(\log|x-z|/|x-z|)$, and at $z\to w$. The first and second terms keep the phase $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$, so their p^+ integral is left undone. They are integrable. In particular, the $\frac{1}{k^+(x-w)^2}$ at $p^+=0$, $x\to w$ becomes $\frac{1}{(x-z)^2}\log\frac{1}{(x-w)^2}$ after $\int dp^+$.

3.4 Row IV

NA

3.5 Row V

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]$$
$$+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[U^{c'e'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]\qquad(3.6)$$

3.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee -k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee -k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}
\tag{3.7}
$$